

Predicting High-Risk Lung Cancer Patients Through Machine Learning-Based Classification Models

Student Name
Department of Computer Engineering
University Name
Turkey
Email: student@example.com

Abstract—Lung cancer is one of the most critical cancer types due to its high mortality rate and the importance of early risk assessment. This study presents a machine learning-based classification system for predicting high-risk lung cancer patients using clinical and diagnostic features. The target variable, *High_Risk*, was derived from cancer stage and performance status indicators. Several supervised learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Gradient Boosting, were implemented and compared. Data preprocessing included categorical encoding, feature scaling, exploratory data analysis, and class imbalance handling using SMOTE. Experimental results showed that Logistic Regression, Random Forest, and Gradient Boosting achieved the highest accuracy scores, approximately 0.71. The developed system demonstrates that machine learning can support clinical risk stratification and assist decision-making in lung cancer analysis.

Index Terms—Machine Learning, Lung Cancer, High-Risk Prediction, Classification, SMOTE, Random Forest, Clinical Decision Support

I. INTRODUCTION

Lung cancer is a major health problem worldwide and requires early diagnosis, accurate classification, and effective treatment planning. In clinical practice, identifying high-risk patients is important because these patients may require closer monitoring, faster intervention, and more detailed treatment planning. Traditional risk assessment is generally based on clinical experience, cancer stage, performance status, and laboratory results.

Machine learning provides an opportunity to analyze many clinical features at the same time and detect patterns that may not be easily observed manually. In this project, a complete machine learning pipeline was developed to classify lung cancer patients as high-risk or non-high-risk.

The main objective of this study is to predict the *High_Risk* class using supervised machine learning models and compare their performances.

II. DATASET DESCRIPTION

The dataset used in this study is named *lung_cancer_data.csv*. It includes demographic, clinical, tumor-related, treatment-related, and laboratory features. Some of the main attributes include age, gender, smoking history, tumor size, tumor location, cancer stage, treatment type, survival months, comorbidities, performance status,

blood pressure, hemoglobin level, white blood cell count, platelet count, albumin level, LDH level, calcium level, glucose level, sodium level, potassium level, and smoking pack years.

The target variable was created as *High_Risk*. A patient was labeled as high-risk if the performance status was greater than or equal to 3 or if the cancer stage was in advanced categories.

$$High_Risk = \begin{cases} 1, & Performance_Status \geq 3 \text{ or } Stage \in \{2, 3\} \\ 0, & otherwise \end{cases} \quad (1)$$

This target definition allows the classification system to focus on patients with advanced disease indicators or poor performance status.

III. EXPLORATORY DATA ANALYSIS

Exploratory data analysis was performed to understand the structure of the dataset, feature distributions, target imbalance, and relationships between variables.

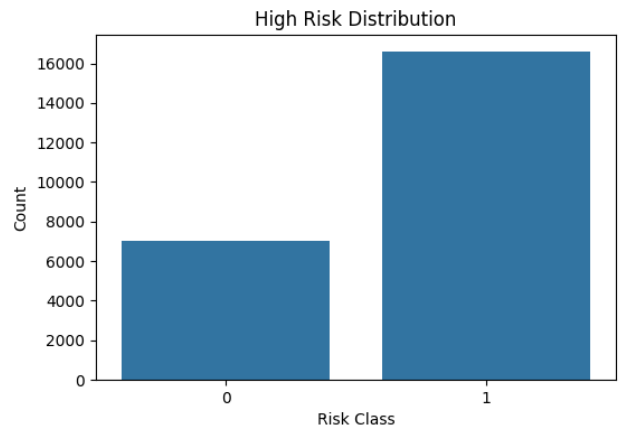


Fig. 1. Distribution of the *High_Risk* target variable.

As shown in Fig. 1, the dataset contains more high-risk patients than non-high-risk patients. This indicates class imbalance, which may affect model performance.

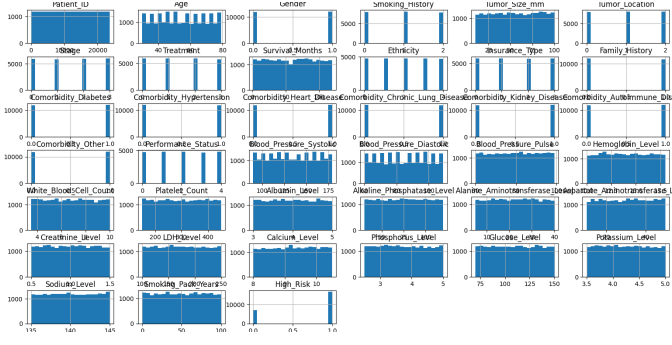


Fig. 2. Feature distribution of all dataset columns.

The distribution of features was analyzed to understand numerical ranges and categorical variable behavior. This step was important before applying scaling and encoding operations.

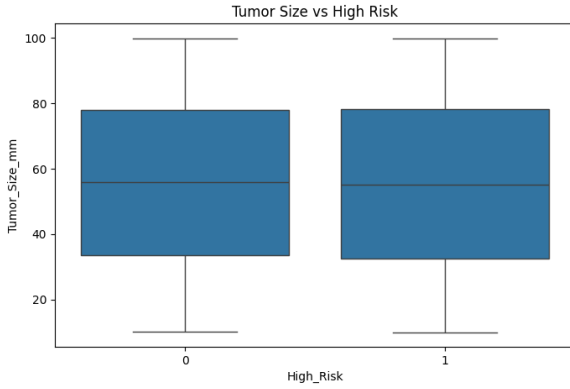


Fig. 3. Tumor size distribution according to High_Risk classes.

Fig. 3 shows the relationship between tumor size and the High_Risk class. The distributions of both classes are similar, which suggests that tumor size alone is not sufficient for reliable risk classification.

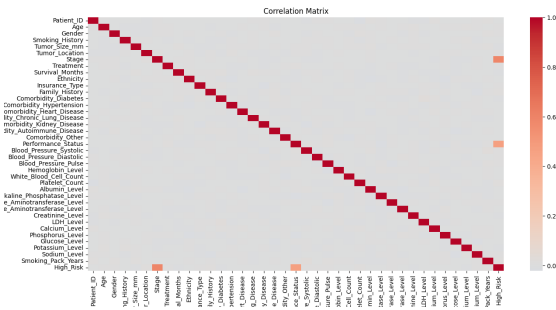


Fig. 4. Correlation heatmap of dataset variables.

The correlation matrix was used to examine relationships between numerical and encoded categorical features. The strongest visible relationships with High_Risk were related to stage and performance status, which is expected because these variables were used in the target definition.

IV. METHODOLOGY

The methodology consists of data preprocessing, target creation, feature preparation, model training, model evaluation, class imbalance handling, and feature importance analysis.

A. Data Preprocessing

First, the dataset was loaded and checked for missing values. Categorical features were converted into numerical form using encoding methods. Numerical features were scaled using standardization. This was especially important for distance-based models such as K-Nearest Neighbors.

The dataset was split into training and testing subsets. The training set was used to fit the models, while the testing set was used to evaluate generalization performance.

B. Class Imbalance Handling

The High_Risk target variable was imbalanced. Therefore, SMOTE was applied to the training data. SMOTE generates synthetic samples for the minority class and helps the model learn a more balanced decision boundary.

C. Machine Learning Models

Five supervised machine learning algorithms were implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors
- Gradient Boosting Classifier

These models were selected because they represent different learning approaches such as linear classification, tree-based learning, ensemble learning, distance-based classification, and boosting.

V. EXPERIMENTAL RESULTS

The models were evaluated using accuracy and confusion matrices. The main objective was to compare the classification performance of different algorithms on the High_Risk prediction task.

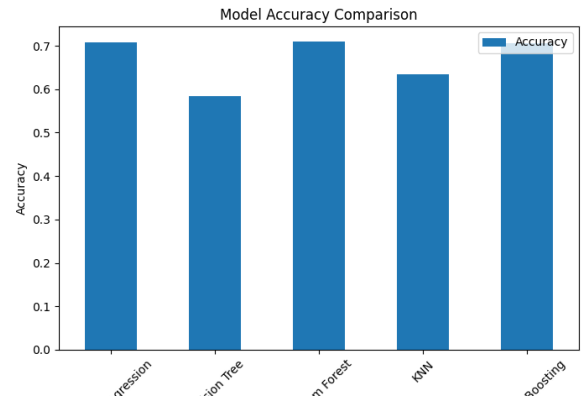


Fig. 5. Accuracy comparison of machine learning models.

The accuracy comparison showed that Logistic Regression, Random Forest, and Gradient Boosting achieved the highest performance with approximately 0.71 accuracy. Decision Tree achieved the lowest accuracy, approximately 0.58, while KNN reached approximately 0.64.

TABLE I
MODEL ACCURACY COMPARISON

Model	Accuracy
Logistic Regression	0.71
Decision Tree	0.58
Random Forest	0.71
K-Nearest Neighbors	0.64
Gradient Boosting	0.71

A. Confusion Matrix Results

The confusion matrix before applying SMOTE is shown in Fig. 6. The model predicted most samples as high-risk, which indicates the effect of class imbalance.

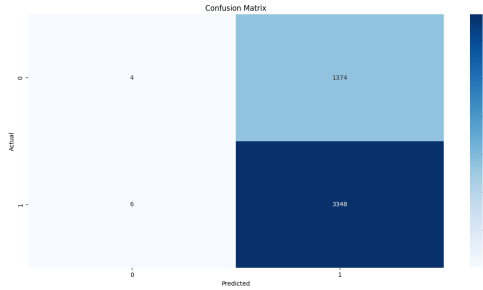


Fig. 6. Confusion matrix before SMOTE.

After applying SMOTE, the confusion matrix changed as shown in Fig. 7. The model still predicted a large number of high-risk samples, but minority class recognition slightly improved.

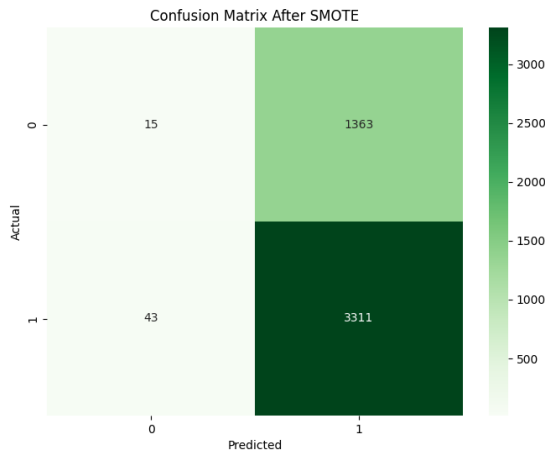


Fig. 7. Confusion matrix after SMOTE.

VI. ROC CURVE ANALYSIS

ROC curve analysis was performed to evaluate classification separability. The ROC-AUC scores of the models were close to 0.50, indicating limited separation capability between the two classes.

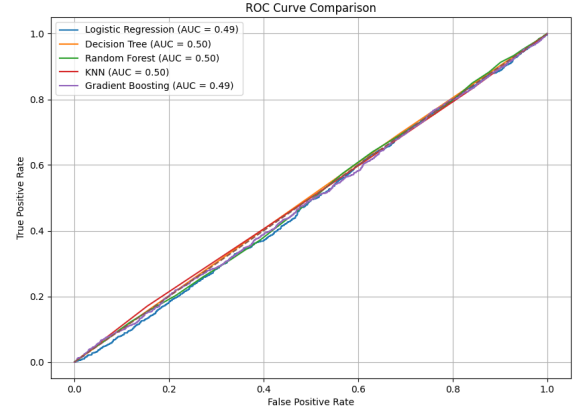


Fig. 8. ROC curve comparison of machine learning models.

Although some models achieved approximately 0.71 accuracy, the ROC-AUC values suggest that the models may be biased toward the majority class. Therefore, accuracy alone is not sufficient for medical classification evaluation.

VII. FEATURE IMPORTANCE AND FEATURE SELECTION

Feature importance analysis was performed using tree-based models. The top features included laboratory and clinical variables such as alkaline phosphatase level, LDH level, phosphorus level, hemoglobin level, creatinine level, sodium level, platelet count, potassium level, smoking pack years, and calcium level.

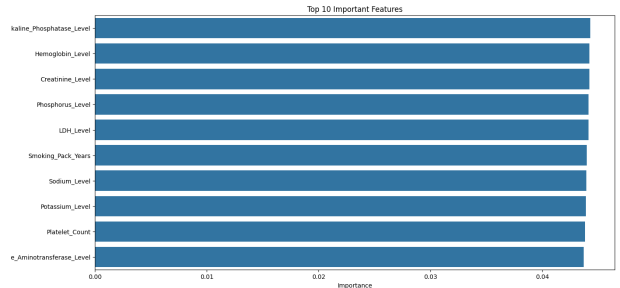


Fig. 9. Top 10 important features in the classification model.

Feature selection was also tested by comparing all features with the top 10 selected features.

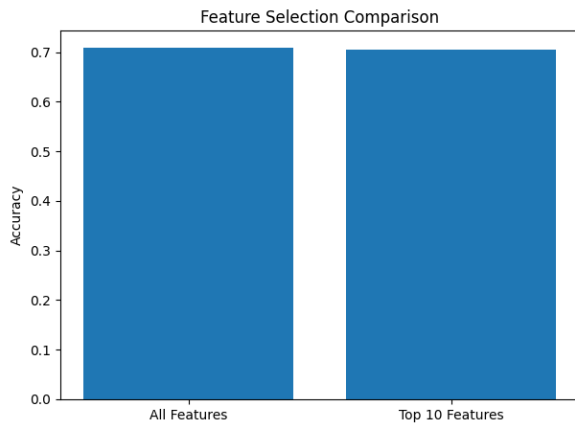


Fig. 10. Accuracy comparison between all features and top 10 selected features.

The results showed that using all features achieved approximately 0.71 accuracy, while using the top 10 features achieved approximately 0.70 accuracy. This indicates that feature selection reduced the number of input variables while maintaining similar performance.

VIII. TRAINING TIME COMPARISON

Training time was also compared to evaluate computational cost.

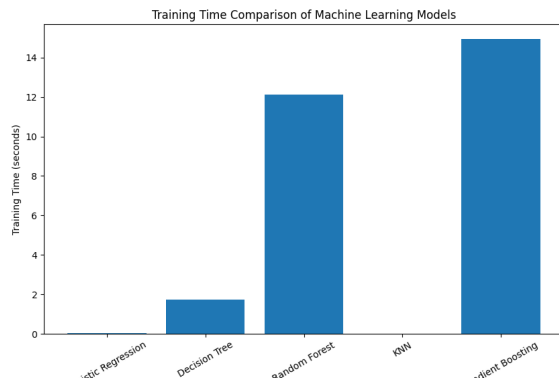


Fig. 11. Training time comparison of machine learning models.

The training time comparison showed that Random Forest and Gradient Boosting required more training time than simpler models. Logistic Regression and KNN had relatively lower training time.

IX. GRAPHICAL USER INTERFACE

A basic graphical user interface was developed to demonstrate how the prediction system can be used with sample patient inputs. The interface allows entering patient values such as age, tumor size, smoking pack years, and survival months.

Fig. 12. Graphical user interface of the lung cancer high-risk prediction system.

The GUI demonstrates the usability of the model as a simple decision-support prototype. However, the full model requires all selected dataset features for complete prediction.

X. DISCUSSION

The results show that machine learning models can be used to classify high-risk lung cancer patients based on clinical and diagnostic features. Logistic Regression, Random Forest, and Gradient Boosting produced the best accuracy scores among the tested models. However, ROC-AUC values close to 0.50 show that the models have limited discriminative ability.

This result is important because it shows that a high accuracy score may not always indicate a strong medical classification model, especially when the target class is imbalanced. Since the dataset contains more high-risk samples, models may achieve acceptable accuracy by predicting the majority class more frequently.

SMOTE was applied to reduce imbalance, but the confusion matrix results show that further improvement is needed. Future studies may include hyperparameter tuning, additional feature engineering, alternative sampling techniques, and validation using real clinical datasets.

XI. CONCLUSION

This study presented a machine learning-based lung cancer high-risk prediction system. A High_Risk target variable was created using cancer stage and performance status. Several supervised learning models were trained and compared. Logistic Regression, Random Forest, and Gradient Boosting achieved the highest accuracy values, approximately 0.71.

Exploratory data analysis, correlation analysis, feature importance, feature selection, confusion matrices, ROC curves, and training time comparisons were included in the evaluation process. A simple GUI prototype was also developed to demonstrate practical usage.

The project shows that machine learning can support risk classification in lung cancer analysis. However, due to limited

ROC-AUC performance, the system should be considered as a decision-support prototype rather than a clinical diagnosis tool.

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